equicenta® CTM

Clinical Evidence Booklet

Regenerative musculoskeletal tissue support for horses — peer-reviewed in Frontiers in Veterinary Science, 2026.

"Horse First, Sport Second."

Equine Performance Labs' guiding principle for equicenta® CTM development.

PRIMARY PUBLICATION

Bertone AL, Davern AJ, Sutter WW, et al. Cryopreserved umbilical allograft (equicenta® CTM) improves clinical outcomes compared with clinician-selected standard care in performance horses: a pragmatic prospective controlled trial. Front. Vet. Sci. 2026. doi:10.3389/fvets.2026.1833933

Study at a glance

A pragmatic prospective controlled trial in performance horses — peer-reviewed in Frontiers in Veterinary Science, 2026.

Peer-reviewed clinical trial data

Study registry: IACUC ETCR-2024-0329 · AVMA VCT #26005934

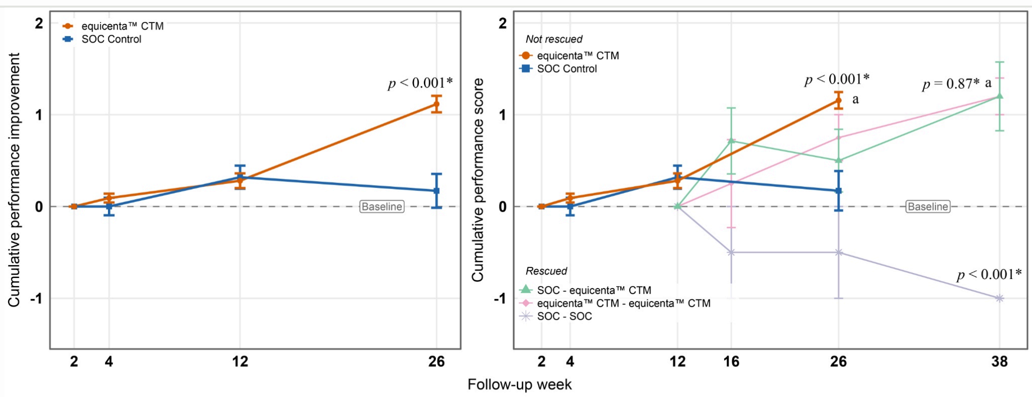
By the numbers

138	195	8	26	1.9:1
HORSES	CONDITIONS	PRACTICES	WEEKS	ALLOCATION
performance cohort	joint · ligament · tendon	16–24 investigating vets	performance follow-up	eCTM : SOC

Abstract (plain language)

In a multicenter controlled trial, a single injection of equicenta® CTM — a cryopreserved equine umbilical cord tissue allograft — was compared with clinician-selected standard-of-care biologics. Horses receiving eCTM showed greater improvement in pain, swelling, and lameness through 12 weeks, higher odds of returning to prior performance at 26 weeks, and a favorable injection-reaction profile. Results are peer-reviewed; the product remains investigational.

Return to — or exceeding — prior performance through 26 weeks (eCTM vs SOC)



	Wk 2	Wk 4	Wk 12	Wk 26
eCTM	86	87	89	86
SOC	45	45	47	41

	Wk 2	Wk 4	Wk 12	Wk 16	Wk 26	Wk 38
eCTM	86	87	89	-	83	-
SOC	45	45	47	-	35	-
SOC - eCTM	-	-	9	7	6	5
eCTM - eCTM	-	-	6	4	4	5
SOC - SOC	-	-	2	2	2	2

Figure 3. Cumulative performance improvement over follow-up: equicenta® CTM (orange) vs clinician-selected SOC control (blue). ITT 26-week effect size Cohen's d = 2.23, p<0.001. Error bars = SEM.

KEY RESULT

26-week performance effect size vs. reference thresholds — small (0.2), medium (0.5), large (0.8); observed **d = 2.23**, far beyond the large-effect threshold.

REP NAVIGATION · WALK THE PAPER IN ORDER

Walk the manuscript, front to back

Every table and figure in the paper's own order, mapped to the page where you can show it. Flip here first, then take the vet through in sequence.

HOW TO USE THIS PAGE

Walk a vet through the paper in its **published order**. **In paper** = the page in Bertone et al. 2026 (36-pp PDF) to cite; **In booklet** = where it lives here. Every page has a "What to tell the vet" box.

MAIN PAPER — IN ORDER

SUPPLEMENTARY — IN ORDER

SEQUENCE TIP

If a vet wants the rigorous walk: Methods → CONSORT (Fig 1) → baseline tables (1–3) → clinical (Fig 2) → performance (Fig 3, Table 4) → safety → owner/imaging (Figs 4–5) → kinematics (Fig 6) → subgroups (Fig 7) → class head-to-head (Fig 8) → supplement.

Source map: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. doi:10.3389/fvets.2026.1833933

SECTION 01 · PRODUCT & INDICATION CONTEXT

equicenta® CTM — cryopreserved umbilical cord tissue allograft

A ready-to-use orthobiologic studied for joint, tendon, and ligament conditions in performance horses.

What it is

equicenta® CTM is a cryopreserved equine umbilical cord–derived connective tissue matrix (CTM) allograft for local administration. In the published trial, every horse in the eCTM arm received a single injection as studied — without a multi-step autologous processing series in that arm. The comparator was clinician-selected standard of care (SOC) biologics used in real-world practice — a pragmatic design intended to reflect field decision-making rather than an artificial single-comparator lab model.

Product specification

TISSUE SOURCE	Equine umbilical cord–derived connective tissue matrix (allograft)
PRESERVATION	Cryopreserved; ready-to-use, no processing series required at time of injection
REGIMEN STUDIED	Single local injection per treated condition (as administered in the trial)
COMPARATOR	Clinician-selected standard-of-care biologic (real-world practice choice)
CONDITIONS STUDIED	Joint, tendon, and ligament conditions — 195 conditions across 138 horses
REGULATORY STATUS	Investigational — not approved or cleared by U.S. FDA Center for Veterinary Medicine

KEY TAKEAWAY

A single studied injection, compared in a peer-reviewed trial against the standard-of-care biologics already in use. Findings are intended to inform discussion between veterinarian and owner — not to substitute for clinical judgment.

SCOPE OF CLAIMS

equicenta® CTM has not been shown to cure disease, and absolute safety has not been established. Exploratory analyses in this booklet are identified as exploratory throughout.

WHAT TO TELL THE VET

One product, one injection, off the shelf. No blood draw, no spin-down, no staged autologous series at injection time. When a vet asks 'what is it,' say: cryopreserved equine umbilical-cord tissue, ready to use, studied as a single local injection per condition. It's investigational — don't state or imply FDA approval. MANUSCRIPT · P2–3 (INTRODUCTION & METHODS)

SECTION 02 · STUDY DESIGN

First controlled trial of an equine umbilical allograft

Prospective, multicenter, quasi-randomized, open-label, controlled pragmatic trial.

138	195	8	1.9:1	100%	98%
HORSES	CONDITIONS	U.S. PRACTICES	ECTM : SOC	WK12 COMPLETE	WK26 RETAINED
enrolled	treated	16–24 investigating vets	91 : 47	138/138	135/138

METHODS ESSENTIALS

- Clinician-selected SOC biologic comparator (real-world care).
- Quasi-randomized within practice to a pre-specified allocation target.
- Primary clinical window: Weeks 0, 2, 4, 12 (lameness, pain, swelling).
- Performance and owner-reported outcomes through Week 26.
- Blinded secondary assessments: Tissue Remodeling Score ultrasound; Sleip kinematics.
- CONSORT-guided reporting; third-party biostatistics.
- IACUC ETCR-2024-0329 · AVMA VCT #26005934.

WHY THE DESIGN MATTERS

Case series can show that some horses improved. A controlled trial asks a harder question: did this product outperform the biologics already used in practice, under the same follow-up rules?

Near-complete follow-up (100% at the Week-12 primary clinical endpoint; 98% at Week 26) reduces the risk that results reflect only the easiest cases to keep in the study.

Enrolled n=138 horses · 195 conditions → **Allocated** eCTM n=91 · SOC n=47 → **Week 12 analyzed** 138/138 (100%) → **Week 26 analyzed** 135/138 (98%)

HEADLINE RESULT (PUBLICATION)

eCTM outperformed SOC across clinical outcomes by 12 weeks ($p<0.001$) and showed a large performance advantage at 26 weeks (Cohen's $d = 2.23$, $p<0.001$), with higher odds of return to prior performance (OR 2.72, $p<.006$).

WHAT TO TELL THE VET

Lead with the design, not the result. This is the first **controlled** trial of an equine umbilical allograft — 138 horses, 8 practices, compared head-to-head against the biologics vets already reach for. Near-complete follow-up (100% at Week 12, 98% at Week 26) means the numbers aren't just the easy cases that stuck around. MANUSCRIPT · P3–7 (METHODS)

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Results §3.1; Methods §2.1–2.9; Figure 1. doi:10.3389/fvets.2026.1833933

SECTION 03 · PUBLICATION CONCLUSIONS

Overarching conclusion from the peer-reviewed paper

Verbatim from the peer-reviewed publication (§5 Summary and conclusion).

> *In this pragmatic setting, a single off-the-shelf injection of eCTM was associated with earlier, greater, and more durable improvements in clinical and performance outcomes than clinician-selected non-steroidal SOC. Overall, these positive findings suggest eCTM may represent a consistent, single-dose alternative to clinician SOC approaches for selected musculoskeletal conditions in performance horses.*

Five evidence pillars

01 **Clinical**	Greater cumulative improvement in pain, swelling, and lameness through Week 12 (p<0.001). Separation by Week 2 (pain/swelling) and Week 4 (lameness).
02 **Performance**	Large ITT performance effect at 26 weeks (d=2.23, p<0.001). Return to prior performance OR 2.72 (p<.006).
03 **Safety**	Lower injection-reaction odds vs SOC (vet 4.8:1; central 5.9:1). Per-injection reactions 3.4% eCTM vs up to 10.5% SOC. No eCTM product-related SAEs reported.
04 **Owner view**	Week-26 owner scores favored eCTM for body condition, coat, overall health, and quality-of-life index.
05 **Structure***	Exploratory ultrasound: Responding tissue-remodeling score 36% eCTM vs 0% SOC ligaments (p=0.013).

*EXPLORATORY QUALIFIER

Imaging and some subgroup contrasts are exploratory analyses. Controlled clinical outcomes, performance odds, and safety are the primary, pre-specified evidence in this study.

WHAT TO TELL THE VET

This is your one-slide summary if a vet is short on time. Walk the five pillars top to bottom: clinical, performance, safety, owner view, structure. Flag pillar 5 (imaging) as exploratory out loud — vets trust reps who name the limits. MANUSCRIPT · ABSTRACT P1–2

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Discussion; Results §3.4. doi:10.3389/fvets.2026.1833933

SECTION 04 · CLINICAL EFFECTIVENESS

Earlier separation — sustained through the Week-12 clinical window

Lameness, pain, and swelling versus clinician-selected standard of care.

p<0.001	Wk 2	Wk 4	3:1	>1.6:1
ALL CLINICAL OUTCOMES	PAIN & SWELLING	LAMENESS	PAIN-FREE ODDS	SOUNDNESS ODDS
through Week 12	first separation	significant	Week 12 · p<0.001	directional · p<0.06

1 **Faster and greater cumulative improvement.** eCTM produced greater cumulative improvement in pain, swelling, and lameness than SOC. Significance by Week 2 for pain (p<0.02) and swelling (p<0.02), by Week 4 for lameness (p<0.01), sustained through 12 weeks (p<0.001 all).

2 **Complete resolution odds.** Odds of score-zero at Week 12 favored eCTM: pain-free 3:1 (p<0.001); soundness (lameness 0) >1.6:1 (near-threshold — communicate directionally).

3 **Clinical window boundary.** Clinical scores are reported through Week 12. Performance is a separate 26-week endpoint and must not be mixed into clinical score trajectories.

STATISTICAL SIGNIFICANCE BY MEASURE/TIMEPOINT

Bar length = −log10(p); longer bar indicates stronger statistical significance.

IN PLAIN TERMS

Horses treated with eCTM typically started looking more comfortable within the first recheck window and remained ahead on scored clinical measures through three months.

WHAT "STANDARD OF CARE" MEANT HERE

The comparator arm was not a placebo. Each SOC joint received a clinician-selected, **non-steroidal** therapy — most often autologous blood-derived orthobiologics, and also amnion allograft, hyaluronic acid, or polyacrylamide gels — so eCTM was measured against real-world best practice, not against no treatment.

HEAD-TO-HEAD VS. HEMODERIVATIVES

A dedicated head-to-head contrast against the largest SOC class — blood-derived orthobiologics (APS, PRP, ACS, PPP) — with the published Figure 8 trajectories appears in **Section 09**.

WHAT TO TELL THE VET

The story here is **speed**: horses looked more comfortable by the first recheck (Week 2 for pain and swelling), and stayed ahead through 12 weeks. Point out the comparator was real orthobiologics, not placebo — so this is 'better than what you already use,' not 'better than nothing.' MANUSCRIPT · §3.4.1.1 P12; FIGURE 2 P14

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Results §3.4.1.1. doi:10.3389/fvets.2026.1833933

SECTION 05 · PERFORMANCE & RETURN-TO-WORK

Return to — or exceeding — prior performance at 26 weeks

A pre-specified performance endpoint from the published trial.

OR 2.72	d = 2.23	79%	21%
RETURN-TO-PERFORMANCE	EFFECT SIZE (ITT)	ECTM SHARE OF SCORE-4	SOC SHARE OF SCORE-4
p<.006	p<0.001	horses reaching Score-4	horses reaching Score-4

COMPOSITION OF HORSES ACHIEVING SCORE-4 (RETURN TO OR EXCEED PRIOR PERFORMANCE)

equicenta® CTM (single injection) 79%

Standard of care (clinician-selected) 21%

ADDITIONAL PERFORMANCE FINDINGS

- Mean performance was higher for eCTM at every evaluation point in the study window.
- ITT performance advantage at 26 weeks was large (Cohen's d = 2.23, p<0.001).
- Advantage preserved in intention-to-treat and per-protocol analyses.

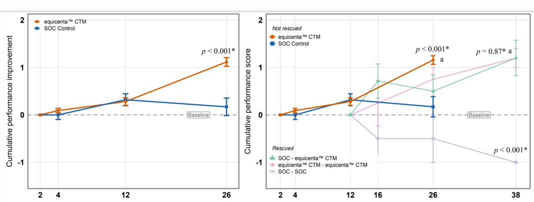
IN PLAIN TERMS

Horses that received eCTM had nearly three times the odds of getting back to their prior performance level compared with horses on standard biologics.

READING THIS STATISTIC CORRECTLY

Per the manuscript, a greater proportion of horses achieving a performance score of 4 (return to or exceeding prior performance) were eCTM (79%) than SOC (21%). This describes the composition of horses reaching Score-4 — not a within-arm success rate. The manuscript reports the treatment effect as odds of returning to or exceeding prior performance (OR 2.72, p<0.006) and ITT effect size (Cohen's d = 2.23, p<0.001); it does not state a per-arm responder percentage.

Performance trajectory and rescue crossover (eCTM vs SOC)



	Wk 2	Wk 4	Wk 12	Wk 26
eCTM	86	87	89	86
SOC	45	45	47	41

	Wk 2	Wk 4	Wk 12	Wk 16	Wk 26	Wk 38
eCTM	86	87	89	-	83	-
SOC	45	45	47	-	35	-
SOC - eCTM	-	-	9	7	6	5
eCTM - eCTM	-	-	6	4	4	5
SOC - SOC	-	-	2	2	2	2

Figure 3. Left: cumulative performance improvement, eCTM (orange) vs SOC (blue), through Week 26 (ITT d=2.23, p<0.001). Right: rescue crossover (exploratory/secondary) — SOC→eCTM recovered to levels indistinguishable from first-line eCTM (p=0.87); SOC→SOC declined (p<0.001). Error bars = SEM.

WHAT TO TELL THE VET

Anchor on the two manuscript numbers — OR 2.72 (nearly 3× the odds of returning to prior performance) and Cohen's d = 2.23 (very large effect).

Say the 79/21 correctly: of horses that returned to prior performance, most were on eCTM. MANUSCRIPT · §3.4.2.1 P14; FIGURE 3 P15

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Results §3.4.2.1; Figure 3. doi:10.3389/fvets.2026.1833933

SECTION 06 · SAFETY

Injection reactions and injection-site inflammation

Head-to-head observation versus clinician-selected standard of care in the same trial.

3.4%	10.5%	4.8:1	5.9:1	0/148
ECTM REACTIONS	SOC REACTIONS	VET-REPORTED ODDS	CENTRAL REVIEW ODDS	ISI SCORE >0
5/148 injections	8/76 · central review	SOC higher · p<0.04	SOC higher · p=0.002	eCTM injections

ECTM ARM

- 5 reactions across 148 injections (3.4% per-injection).
- 0/148 injections with injection-site inflammation (0–4 scale) at weeks 2, 4, and 12.
- 0/148 infections; no vet-reported product-related serious adverse events.
- Observed reactions were mild, self-limited, and early.

SOC ARM (CONTEXT)

- 8 reactions / 76 injections under central review (10.5%).
- 2/76 serious intra-articular reactions in SOC; none with eCTM.
- Advantage held on vet report and blinded central adjudication.
- PAAG contributed to excess SOC joint reactions; excluding PAAG, SOC odds remained higher (3.4:1, p<0.03).

IN PLAIN TERMS

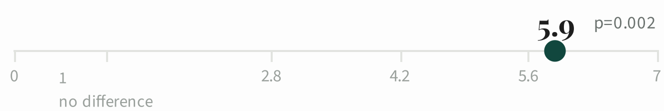
Standard biologics in this study were associated with several times more injection reactions. No serious product-related events were attributed to eCTM.

INVESTIGATIONAL STATUS

equicenta® CTM is investigational. No side effects or absolute safety have been established, and individual results may vary.

PER-INJECTION REACTION RATE

SOC-VS-ECTM REACTION ODDS (CENTRAL REVIEW)



WHAT TO TELL THE VET

Vets buy on safety. Fewer injection reactions on eCTM than SOC — 3.4% per injection vs up to 10.5%, and no product-related serious events on eCTM. Stay honest: 'absolute safety isn't established, but every reaction count favored eCTM in this trial.'

MANUSCRIPT · §3.4.1.2 P14; SUPPL TABLE S5 P33

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Results §3.4.1.2. doi:10.3389/fvets.2026.1833933

SECTION 07 · OWNER OUTCOMES & STRUCTURAL SIGNAL

What owners reported — and what imaging explored

Secondary owner endpoints at Week 26; exploratory tissue remodeling on ultrasound.

A. OWNER-REPORTED OUTCOMES (WEEK 26)

p<.004	p<.029	p<.001	p<.03
BODY CONDITION	HAIR COAT	OVERALL HEALTH	QOL INDEX
0–10 owner scale	0–10 owner scale	0–10 owner scale	0–70 paper sum

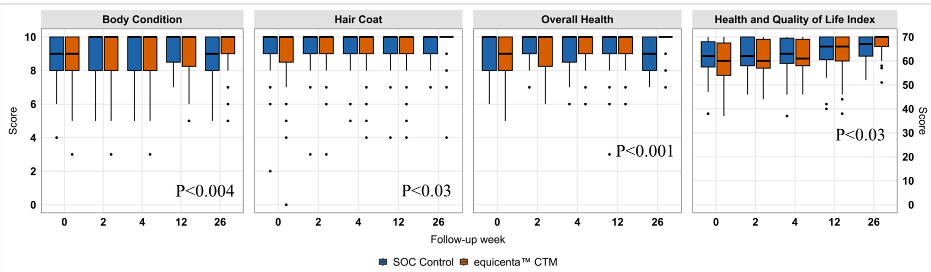


Figure 4. Owner-reported secondaries at Week 26 (eCTM orange vs SOC blue). Direction of improvement aligned with clinician-scored outcomes. QOL index uses a 0–70 scale, reported as such (not converted to 0–10).

B. STRUCTURAL HEALING — EXPLORATORY TRS

36%	0%	d=0.98	d=1.25
ECTM RESPONDING TRS	SOC RESPONDING TRS	OVERALL LIGAMENTS	PROXIMAL SUSPENSORY
8/22 ligaments	0/11 ligaments	p=0.013	subclass

Final tissue remodeling score (TRS)

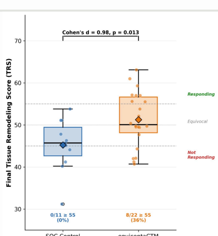
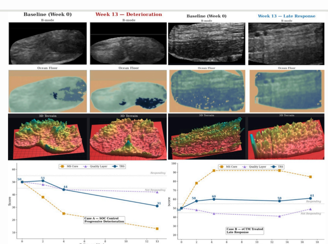


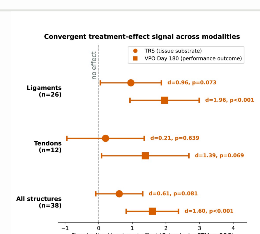
Figure 5. Final TRS by arm: eCTM (orange) vs SOC (blue). Cohen's d=0.98, p=0.013; responding (≥55) 8/22 eCTM vs 0/11 SOC. Exploratory.

TRS ultrasound case examples (baseline → Week 13)



Suppl. Fig. S1. Case A (SOC): deterioration, TRS 50→31. Case B (eCTM): late response, TRS 50→61 (crossing responding threshold). Exploratory.

Convergent treatment-effect signal



Suppl. Fig. S2. TRS vs performance effect sizes across ligaments, tendons, all structures. Exploratory; tendon estimate small (n=13).

WHAT TO TELL THE VET

Two audiences on one page. Owner-reported scores (body condition, coat, overall health, quality of life) give the barn-side language owners care about. The ultrasound tissue-remodeling signal is **exploratory** — use it to show the tissue may be rebuilding, but always label it exploratory. MANUSCRIPT · §3.4.2.2 P14 (OWNER); §3.4.2.4 P16 (TRS)

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. §3.4.2.2 Fig 4; §3.4.2.4 Fig 5; Suppl. Figs S1–S2. doi:10.3389/fvets.2026.1833933

SECTION 08 · RIGOR, FOLLOW-UP & ONSET

Study integrity and when improvement separates

Design safeguards documented in the peer-reviewed publication.

A. RIGOR & FOLLOW-UP

138/138	135/138	Blinded	CONSORT
WEEK 12	WEEK 26	SECONDARIES	REPORTING
100% primary	98% retained	US · kinematics · stats	pre-specified

- Independent third-party biostatistics; analysts blinded where specified.
- Standardized AAEP lameness and validated 0–4 pain/swelling scores.
- Bias safeguards: eligibility, consecutive enrollment, stratified allocation, consensus training.
- Study rigor is documented ahead of the efficacy findings in the manuscript.
- The DOI and trial IDs are provided for independent verification.
- Complete follow-up supports both intention-to-treat and per-protocol interpretation.

B. ONSET TIMELINE (CLINICAL WINDOW)

Baseline	Week 2	Week 4	Week 12	Week 26
Enroll	Pain/swelling	Lameness	All clinical	Performance
	$p<0.02$	$p<0.01$	$p<0.001$	(separate)

IN PLAIN TERMS

Improvement is not only a six-month story. Clinical separation begins at the early rechecks and is measured rigorously through Week 12; return-to-work is then confirmed at Week 26.

CONSECUTIVE	STRATIFIED	INDEPENDENT	CONSENSUS
Enrollment, not selective	Allocation by practice	Third-party biostatistics	Scorer training

WHAT TO TELL THE VET

Use this when a skeptical vet asks 'how good is the study, really?' Consecutive enrollment, stratified allocation, independent biostatistics, trained consensus scorers. Then walk the onset timeline: separation starts at Week 2, holds through 12, performance confirmed at 26. **MANUSCRIPT · METHODS P3–7; FIGURE 1 P9**

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. §3.1; Fig 1; §3.4.1.1 Fig 2. doi:10.3389/fvets.2026.1833933

SECTION 09 · HEAD-TO-HEAD VS. HEMODERIVATIVES

Tested against the prevailing biologic paradigm

A predefined post hoc contrast of eCTM against pooled autologous blood-derived products — the largest treatment class in the standard-of-care arm.

63.2%	n=43	p<0.04	p<0.003	p<0.001
OF THE SOC ARM	HEMODERIVATIVE	LAMENESS	PAIN	SWELLING
were hemoderivatives	conditions (SOC)	Weeks 4 & 12	on flexion	effusion

1 **The comparator that matters.** Autologous blood-derived orthobiologics — APS, PRP, ACS, and PPP — are the dominant intra-articular biologics in equine practice. In this pragmatic trial they formed the largest single subgroup of the clinician-selected SOC arm (n=43; 63.2%), so this contrast pits eCTM against the prevailing standard.

2 **eCTM improved more, across timepoints.** In the pooled hemoderivative contrast, eCTM showed greater cumulative improvement at multiple timepoints (range p<0.04 to p<0.001), most clearly on lameness (Weeks 4 & 12, p<0.04), pain (p<0.003), and swelling (p<0.001).

3 **Single dose vs. the class.** Each eCTM joint received one 1.5 mL intra-articular injection of cryopreserved umbilical-derived protein/microparticle suspension — compared against the range of blood-derived products chosen by experienced clinicians in real-world practice.

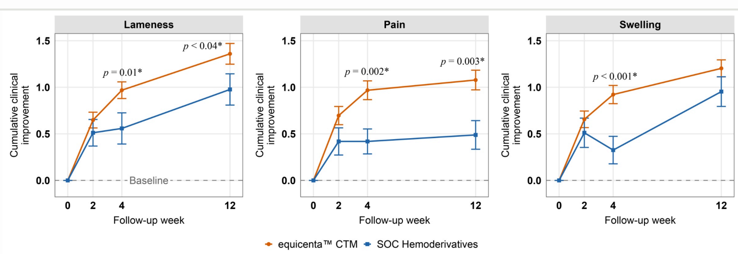
HOW TO READ THIS CONTRAST

This is a **predefined post hoc exploratory subgroup analysis**, one of two clinically sensible SOC contrasts identified by the authors. It uses the pooled standard-of-care variance estimate. Report it as supportive, class-level evidence — not as a primary registrational endpoint.

IN PLAIN TERMS

Against the blood-derived products that make up most of everyday joint therapy, a single off-the-shelf eCTM injection produced greater improvement in lameness, pain, and swelling through the 12-week clinical window.

Cumulative improvement from baseline — eCTM vs. pooled hemoderivatives (lameness, pain, swelling)



	Wk 0	Wk 2	Wk 4	Wk 12
equicenta™ CTM	128	125	128	128
SOC Hemoderivatives	43	43	43	43

	Wk 0	Wk 2	Wk 4	Wk 12
equicenta™ CTM	128	125	126	128
SOC Hemoderivatives	43	43	43	43

	Wk 0	Wk 2	Wk 4	Wk 12
equicenta™ CTM	128	125	128	128
SOC Hemoderivatives	43	43	43	43

SOC Hemoderivatives include APS, PRP, ACS, and PPP.

Figure 8. Sub-class analysis for hemoderivatives. Cumulative clinical trajectories from baseline (mean ± SEM) were greater for eCTM (orange) than the SOC subclass of hemoderivatives (blue) at most timepoints over 12 weeks — particularly lameness (Weeks 4 & 12; p<0.04) and pain (p<0.003). Hemoderivatives = APS, PRP, ACS, and PPP. Predefined exploratory subgroup analysis.

WHAT TO TELL THE VET

Your APS conversation: eCTM tracked ahead of the blood-derived products (APS/PRP/ACS/PPP). Keep it exploratory — the primary claim is eCTM vs all SOC combined. MANUSCRIPT · §3.4.2.7 P17; FIGURE 8 P18; SUPPL FIG S3 P28

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Methods §2.7; Results §3.4 Figure 8; Supplementary Figure S3. doi:10.3389/fvets.2026.1833933

SECTION 09 (CONT.) · HEAD-TO-HEAD VS. HEMODERIVATIVES — MECHANISM & PRACTICE

Why the difference is biologically plausible

Duration of effect, injection burden, and safety — the practice case behind the numbers.

DURATION OF PRIOR EVIDENCE	Controlled evidence for blood-derived orthobiologics extends only to about **2 weeks** (one controlled study of APS), with longer-term outcomes unknown. Their soluble proteins are expected to be rapidly absorbed or degraded, so benefit is anticipated to be short-lived.
ECTM MECHANISM	Umbilical allograft (eCTM) delivers fetal anabolic and reparative proteins, in part as a biodegradable connective tissue that **releases proteins over time** — a plausible mechanism for the more sustained benefit observed relative to the hemoderivative class.
INJECTION BURDEN	Multiple-injection protocols for ACS and PRP are widely used, yet three intra-articular ACS injections did not demonstrate clinical benefit in a controlled study. eCTM was delivered as a single injection (see re-injection rates below).
SAFETY CONSIDERATION	Post-injection infection remains the most frequently reported serious adverse event for PRP and other hemoderivatives. Because any repeated joint injection may raise infection risk, the authors note that justifying multiple injections **should rest on efficacy data or be balanced against options requiring fewer administrations.**
PRACTICE IMPLICATION	Hemoderivatives require blood draw, chairside processing, and often multiple visits. A single, off-the-shelf allograft that outperformed this class on blinded lameness and pain endpoints has meaningful workflow implications.

ADDITIONAL INJECTIONS BY ORIGIN (TABLE 5)

9	11	9
SOC → SOC	SOC → ECTM	ECTM → ECTM
7 prescribed + 2 rescue	rescue crossover	rescue

SOC-assigned horses generated **significantly more** additional injections (20 vs. 9; $p<0.03$), driven largely by protocol-prescribed repeat doses; rescue-injection counts alone did not differ between arms ($p>0.09$).

APS-SPECIFIC CONTRAST (EXPLORATORY · POST HOC)

Why APS

APS was the single most-used hemoderivative (n=22; 51% of hemoderivatives) with prior supportive evidence in equine joint osteoarthritis — so it was chosen for a dedicated contrast.

What was seen

eCTM remained associated with greater improvement at multiple timepoints, primarily in lameness and pain (Supplementary Figure S3).

How to weight it

This APS contrast was **not pre-specified** and was limited by small sample size; treat it as hypothesis-generating, not confirmatory.

EVIDENTIARY NOTE

The pooled-hemoderivative and APS-specific contrasts are both post hoc exploratory subgroup analyses — not primary registrational endpoints. equicenta® CTM remains investigational and is not approved or cleared by the U.S. FDA Center for Veterinary Medicine.

WHAT TO TELL THE VET

Keep mechanism simple — a connective-tissue-matrix scaffold in one shot; tie it back to what was measured, not biology the paper doesn't establish. MANUSCRIPT · INTRODUCTION P3; DISCUSSION P18–21

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Discussion §4.1–4.3; Results §3.4; Supplementary Figure S3. doi:10.3389/fvets.2026.1833933

SECTION 10 · OBJECTIVE GAIT & COMPARATIVE CONTEXT

Blinded kinematics and exploratory class comparisons

Supportive analyses, reported at their appropriate evidentiary tier.

A. SLEIP KINEMATICS (EXPLORATORY)

$p<.003$	$p<.05$	$p<.01$	Blinded
PUSH-OFF	IMPACT	PUSH-OFF	ANALYSTS
Week 2	Week 4	Week 12	to treatment

- Reported p-values cover Weeks 2, 4, and 12 only; Week 26 kinematic data were not reported.
- Sensor-based kinematics support the direction of clinical findings and do not replace the primary clinical endpoints.

Cumulative gait asymmetry from baseline (lower = improvement toward symmetry)

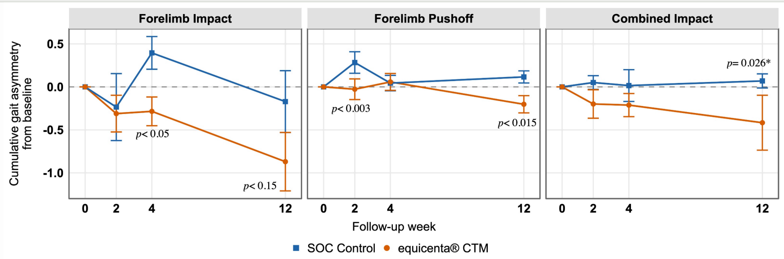


Figure 6. Sleip inertial-sensor kinematics: forelimb impact, forelimb push-off, combined impact. eCTM (orange) vs SOC (blue). Blinded analysts. Exploratory supportive endpoint. Error bars = SEM.

B. RESCUE CROSSOVER (EXPLORATORY / SECONDARY)

Rescue criterion

19 horses (13%) met the pre-defined rescue criterion of no lameness improvement at Week 12 and received rescue injections per protocol.

SOC→eCTM recovery

Horses crossed from SOC to eCTM recovered to levels statistically indistinguishable from first-line eCTM (p=0.87).

Hemoderivative contrast

The full head-to-head vs. blood-derived products (APS, PRP, ACS, PPP), with Figure 8, is presented in Section 09.

EVIDENTIARY NOTE

These comparative and rescue analyses are exploratory or post-hoc unless stated otherwise.

WHAT TO TELL THE VET

Objective gait data (Sleip) backs up what the vets scored by eye — forelimb impact and push-off improved at Weeks 2, 4, 12. Note Week-26 kinematics weren't reported. Kinematics **support** the clinical findings; they don't replace the primary endpoints.

MANUSCRIPT · §3.4.2.5 P16; FIGURE 6 P16

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. §3.4.2.5 Fig 6; §3.4.2.7; rescue §3.4.2.1. doi:10.3389/fvets.2026.1833933

SECTION 11 · DISCUSSION — CLINICAL RELEVANCE

Why the effect is durable — and what it means at chairside

Dr. Bertone and co-investigators interpret the mechanism, timing, and structural signals behind the clinical separation.

Earlier recovery, longer benefit

eCTM-treated conditions showed **earlier divergence and larger cumulative improvements** in lameness, pain, and swelling than SOC, with differences emerging by 2–4 weeks and persisting for at least 12 weeks.

Hemoderivative benefit is expected to be short-lived — soluble proteins are rapidly absorbed or degraded, and one controlled study showed APS benefit out to only 2 weeks. Umbilical allograft instead delivers fetal anabolic and reparative proteins as a biodegradable connective tissue, **releasing proteins over time** — a plausible mechanism for the sustained benefit observed.

Consistent across tissue types

The uniformity of response across joints, tendons, and ligaments was anticipated: many biological processes are shared among musculoskeletal connective tissues, and this is the established rationale for using biologics in these same tissues.

CHAIRSIDE IMPLICATION

Between 12 and 26 weeks, SOC performance gains plateaued or declined — particularly in soft-tissue conditions — whereas eCTM horses continued to improve, suggesting eCTM may help sustain functional recovery.

A single injection vs. repeat protocols

Multiple-injection ACS and PRP protocols are widely used, so SOC received significantly more re-injections than the single eCTM injection. Yet three intra-articular ACS injections did not show clinical benefit in a controlled equine study, and a joint infection occurred after a third injection.

Post-injection infection remains the most frequent serious adverse event for hemoderivatives. Because any repeated joint injection may increase infection risk, the authors note the justification for multiple injections **should rest on efficacy data or be balanced against options requiring fewer administrations.**

Joint pain and performance move together

SOC-treated joints had transient performance improvement not matched by sustained joint-pain relief, then declined toward baseline by 26 weeks. eCTM-treated joints showed **sustained joint-pain improvement with continued performance gains** through 26 weeks.

HYPOTHESIS-GENERATING

As sub-cohort analyses, these suggest eCTM may provide more durable modulation of intra-articular synovial pain than some existing biologics; definitive mechanistic studies would be needed to test this hypothesis.

WHAT TO TELL THE VET

The authors' own read: earlier, greater, more durable improvement than non-steroidal SOC. Use their words — vets respect the discussion section. Don't stretch beyond 'may represent a single-dose alternative for selected conditions.'

MANUSCRIPT · DISCUSSION P18–21

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. §4.1 Clinical effectiveness and durability of response. doi:10.3389/fvets.2026.1833933

SECTION 12 · DISCUSSION — STRUCTURE, SAFETY & DESIGN

Structural signals, a favorable safety profile, and a real-world design

Supportive structural and kinematic evidence, owner-reported outcomes, and the pragmatic trial design — with limitations stated plainly.

A. STRUCTURAL REMODELING & QUANTITATIVE VALIDATION (EXPLORATORY)

Quantitative ultrasound-based **TRS** provided complementary structural evidence of tissue remodeling — particularly in ligament lesions — that may coincide with later performance improvement. Segmentation reproducibility was high, with a **cohort mean Dice coefficient of 0.931**, above published strong intra-rater values.

Tendon TRS correlated significantly with 12-week limb-level lameness and was positively associated with 26-week performance. These findings **do not constitute full validation** but provide exploratory hypothesis generation to justify further work.

Kinematic gait analysis by **blinded analysts** supported the clinical findings, particularly in forelimb lameness: forelimb impact and push-off asymmetry improved more in eCTM horses at several time points. Hindlimb and combined-limb data were highly variable.

EVIDENTIARY TIER

TRS is a novel, not-yet-fully-validated biomarker used here as an exploratory endpoint; kinematics are a supportive endpoint. Both reinforce — but do not replace — the primary clinical outcomes.

B. SAFETY & OWNER-REPORTED OUTCOMES

eCTM was **well tolerated**; product-related adverse events were infrequent, self-limited, and early. SOC treatments were associated with a higher frequency of product-related events overall and in joints specifically, including a small number of serious events. **Two serious joint reactions occurred after polyacrylamide,** one persisting at follow-up. Central review that reclassified events as possibly product-related did not change the safety profile favoring eCTM.

At 26 weeks, owners of eCTM horses reported **higher scores for body condition, coat quality, overall health, and composite quality of life** than SOC owners — suggesting the product will be well received and may reflect subclinical measures of health.

C. PRAGMATIC DESIGN & STATED LIMITATIONS

The trial intentionally reflected real-world practice — 8 high-volume practices, a spectrum of joint and soft-tissue conditions, and an SOC arm spanning treatments in current clinical use — enhancing external validity. The authors state plainly that the study was **open-label and unblinded within practices** (SOC required visible blood collection), mitigated by quasi-randomized allocation, prespecified definitions, standardized scoring, and objective TRS/kinematic measures. Subgroup analyses are **hypothesis-generating and some underpowered;** the trial was powered for the overall eCTM-vs-SOC comparison.

WHAT TO TELL THE VET

This is where limitations live — say them first and you win trust. Open-label design, quasi-randomization, single trial, exploratory imaging. Then pivot: even with those caveats, the pre-specified clinical, performance, and safety endpoints favored eCTM.

MANUSCRIPT · DISCUSSION P18–21

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. §4.2 Structural remodeling; §4.3 Pragmatic design; §4.4 Safety and owner-reported outcomes; §4.5 Limitations. doi:10.3389/fvets.2026.1833933

SECTION 13 · CONCLUSION — CLINICAL IMPLICATIONS

A consistent, single-dose option — where prior biologics fell short

The investigators' clinical implications and summary, in their own framing.

> *In this pragmatic setting, a single off-the-shelf injection of eCTM was associated with earlier, greater, and more durable improvements in clinical and performance outcomes than clinician-selected non-steroidal SOC. Overall, these positive findings suggest eCTM may represent a consistent, single-dose alternative to clinician SOC approaches for selected musculoskeletal conditions in performance horses.*

What the evidence supports

1. **Earlier & greater improvement.** In lameness, pain, and swelling versus standard care currently used in practice.
2. **Sustained performance.** Gains maintained through the 26-week endpoint.
3. **Favorable safety.** Infrequent, self-limited, early product-related events.
4. **Supportive structure & gait.** Consistent exploratory TRS and blinded kinematic signals.
5. **Practical delivery.** Consistent composition, no patient-side processing, single injection.

Where it fits in practice

The authors specifically support consideration of eCTM **in horses with performance-limiting lameness where previous biologic treatments have provided only transient benefit**, as for the horses in this study.

Future directions

Planned work includes stratified cohorts, optimal dosing, and re-treatment strategies, plus mechanistic studies integrating imaging, biomarker, and clinical data to clarify how eCTM influences tissue remodeling and how best to select candidates most likely to benefit.

STUDY CONTEXT

This multicenter, quasi-randomized, open-label controlled trial with long-term follow-up on over a hundred horses in real-world conditions helps fill a knowledge gap; fully blinded RCTs have not been done in horses and carry their own trade-offs in generalizability.

AUTHORS' INDEPENDENCE

All authors were third-party independent collaborators and not employees of EPL. Dr. Alicia Bertone served as an independent external research consultant, and the study's multicenter study director, quality monitor, and third-party adjudicator.

WHAT TO TELL THE VET

Close on the verbatim conclusion and the regulatory line: investigational, not for human use, not for food-producing animals, individual results vary. Leave the clean edition behind so the vet can re-read everything you just walked through. MANUSCRIPT · §5 CONCLUSION P21

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. §4.6 Clinical implications and future directions; §5 Summary and conclusion; Author's note. Product investigational — not FDA CVM approved. doi:10.3389/fvets.2026.1833933

SECTION APX A1 · CONDITIONS TREATED

Every joint, tendon, and ligament condition in the trial

195 conditions across 138 performance horses — broken out by tissue and anatomy.

85	65	28	17	195
JOINTS	LIGAMENTS	TENDONS	OCD LESIONS	TOTAL
arthritis · cysts	desmitis	SDF · DDF	stifle · tarsus · fetlock	138 horses · 8 practices

MANUSCRIPT BASELINE — CONDITIONS BY ARM (BERTONE ET AL. 2026, TABLE 3)

Condition	equicenta® CTM		SOC	
	n	%	n	%
JOINT CONDITIONS	85 TOTAL · 50 ECTM · 35 SOC			
Arthritis / synovitis	44	34.4	32	47.8
› Stifle	10	7.8	8	11.9
› Fetlock	11	8.6	7	10.4
› Coffin joint	6	4.7	7	10.4
› Carpus	6	4.7	2	3.0
› Tarsus	3	2.3	2	3.0
› Hip	0	0.0	1	1.5
› Unspecified joint	8	6.2	5	7.5
Articular cysts	6	4.7	3	4.5
› Stifle	4	3.1	2	3.0
› Fetlock	1	0.8	0	0.0
› Carpus	1	0.8	0	0.0
› Tarsus	0	0.0	1	1.5
OCD LESIONS	17 TOTAL · 9 ECTM · 8 SOC			
OCD	9	7.0	8	11.9
› Stifle	7	5.5	7	10.4
› Tarsus	2	1.6	0	0.0
› Fetlock	0	0.0	1	1.5
TENDON CONDITIONS	28 TOTAL · 22 ECTM · 6 SOC			
Tendonitis	22	17.2	6	9.0
› SDF tendonitis	14	10.9	5	7.5
› DDF tendonitis	8	6.2	1	1.5
LIGAMENT CONDITIONS	65 TOTAL · 47 ECTM · 18 SOC			
Desmitis	47	36.7	18	26.9
› Suspensory branch desmitis	22	17.2	6	9.0
› Proximal suspensory desmitis	17	13.3	8	11.9
› Collateral ligament desmitis	3	2.3	1	1.5
› Check ligament desmitis	2	1.6	0	0.0
› Desmitis (unspecified)	2	1.6	1	1.5
› Annular ligament desmitis	1	0.8	1	1.5
› Manica flexoria desmitis	0	0.0	1	1.5

CASE-MIX AT A GLANCE

Joints made up ~44% of treated conditions, ligaments ~33%, tendons ~14%, and OCD lesions ~9%. Stifle was the most common anatomic site treated overall — appearing across arthritis, articular cysts, and OCD.

SOFT-TISSUE DETAIL

Of the 93 soft-tissue conditions, suspensory ligament injuries dominated: 28 suspensory branch desmitis and 25 proximal suspensory desmitis — 53 of 65 ligament conditions (82%). Tendon conditions split 19 SDF / 9 DDF.

JOINT DETAIL

Of 85 joint conditions, arthritis / synovitis accounted for 76 and articular cysts for 9. Stifle appeared in 38 conditions across arthritis, cysts, and OCD — more than any other anatomic site. Fetlock followed with 20, then coffin joint with 13.

READING THIS TABLE

Percentages are of each arm's total conditions (eCTM = 128; SOC = 67). Sub-rows break out each parent category by anatomy — parent counts, not sub-rows, sum to 195 total conditions across 138 horses.

WHAT TO TELL THE VET

This is Table 3 — the case mix: 85 joint, 65 ligament, 28 tendon, 17 OCD across 195 conditions. Use it to prove breadth, not cherry-picked lesions.

MANUSCRIPT · TABLE 3 P11

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Table 3 — Baseline conditions by arm. doi:10.3389/fvets.2026.1833933

SECTION APX A2 · CASE-MIX · PRIOR TREATMENT

Who these horses were — and what had already been tried

Discipline mix, tissue counts by discipline, and prior-treatment history at enrollment.

DISCIPLINE × CONDITION × WK-26 PERF. (TABLE 4)

Discipline / Arm		n	Lig.	Ten.	Joint	%	Wk26
Dressage	SOC	10	5	1	8	21.3	2.2
	eCTM	9	5	3	3	9.9	3.8
Hunter / Jumper / Event.	SOC	13	6	0	14	27.7	2.5
	eCTM	16	5	5	10	17.6	3.4
Racing	SOC	14	1	3	17	29.8	3.3
	eCTM	36	23	4	26	39.6	3.6
Western	SOC	10	3	2	7	21.3	2.3
	eCTM	20	12	5	11	22.0	3.0
Other*	SOC	0	0	0	0	0.0	—
	eCTM	10	3	5	8	11.0	3.3
GRAND TOTAL							
	SOC	47	15	6	46	100.0	2.7
	eCTM	91	48	22	58	100.0	3.4*

* Wk-26 performance p = 0.004 vs SOC; no significant difference by discipline. Other = competitive gaited, driving, show horses (non hunter/jumper).

PRIOR TREATMENTS (% OF HORSES) — TABLE 1

Class	Treatment	eCTM %	SOC %
Pharma	NSAIDs	56.2	55.6
	Steroids	27.1	22.2
	PSGAG	12.5	25.9
Biologic	APS	20.8	33.3
	PRP	10.4	25.9
	HA	8.3	7.4
	ACS	0.0	7.4
	Amnion allograft	12.5	0.0
	Stem cells	2.1	0.0
	PPP (plt-poor)	2.1	7.4
PT	Shockwave	6.2	0.0
	Rehabilitation	—	—

PRE-TRIAL BURDEN

56% of eCTM and 56% of SOC horses had already tried NSAIDs. One in five eCTM horses had tried APS; one in four SOC horses had tried PRP. These were not first-line cases.

DISCIPLINE READ

Racing was the largest single group in the eCTM arm (n = 36, 40%). Hunter / jumper / eventing was the largest SOC group (n = 13, 28%). Wk-26 performance favored eCTM in every discipline with SOC controls.

WHY THIS MATTERS

Case-mix diversity — 5 disciplines, 5 tissue types, 2 duration bands — is what supports the manuscript's external-validity claim. This was not a single-condition, single-discipline pilot.

WHAT TO TELL THE VET

Table 1 — what these horses were given before enrolling. Many had prior NSAIDs and biologics, so eCTM was tested in real, previously-treated horses, not fresh cases. MANUSCRIPT · TABLE 1 P10

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Tables 1 and 4. doi:10.3389/fvets.2026.1833933

SECTION APX A3 · DIAGNOSTIC WORKUP & CONSORT

Real workups, real cases — a full CONSORT trail

Diagnostic tests recorded at enrollment plus the full participant flow from screening to Week 26.

CONSORT PARTICIPANT FLOW (FIGURE 1)

144 horses
SCREENED FOR ENROLLMENT

→ *6 excluded (screening failure)*

138 horses · 195 conditions
ENROLLED & RANDOMIZED (2:1)

91 horses · 128 cond.
ECTM ARM

47 horses · 67 cond.
SOC ARM

98–100% retention
WK 12 CLINICAL FOLLOW-UP

→ *<2% lost to follow-up*

Return-to-sport scored
WK 26 PERFORMANCE OUTCOME

DIAGNOSTIC TESTS AT ENROLLMENT (TABLE 2)

Test	n eCTM	% eCTM	n SOC	% SOC
Physical examination	62	68.1	37	78.7
Lameness examination	76	83.5	42	89.4
Local anesthesia (blocks)	28	30.8	18	38.3
Radiographs	43	47.3	29	61.7
Ultrasound	63	69.2	23	48.9
CT / MRI / NM	7	7.7	3	6.4
Other	1	1.1	0	0.0
Mean tests / horse	—	**3.0**	—	**3.2**
Total tests recorded	**280**	—	**152**	—

WORKUP RIGOR

Every horse had a formal lameness exam. Half had radiographs, 60% had ultrasound imaging, and ~7% had advanced imaging (CT/MRI/NM). Groups had comparable workups (3.0 vs 3.2 tests / horse).

NOT A CASE SERIES

The 144→138 screen-to-enroll flow, imaging-driven diagnoses, and 98–100% follow-up separate this trial from typical veterinary case-series or single-site reports.

WHAT TO TELL THE VET

Table 2 plus the CONSORT flow (Figure 1). This is your 'the diagnosis was done properly' page — blocks, imaging, and the full screening-to-Week-26 participant flow. Walk the flow boxes left to right. MANUSCRIPT · TABLE 2 P11; FIGURE 1 P9

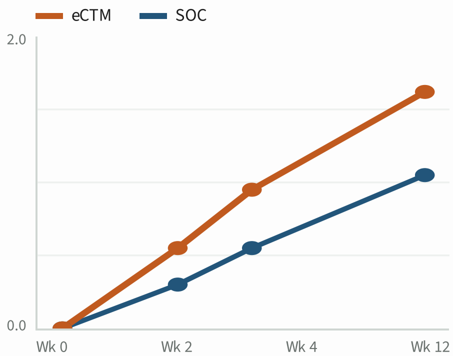
Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Table 2 and Figure 1. doi:10.3389/fvets.2026.1833933

SECTION APX A4 · CLINICAL TRAJECTORIES WK 0–12

The improvement gap opened at Week 2 and never closed

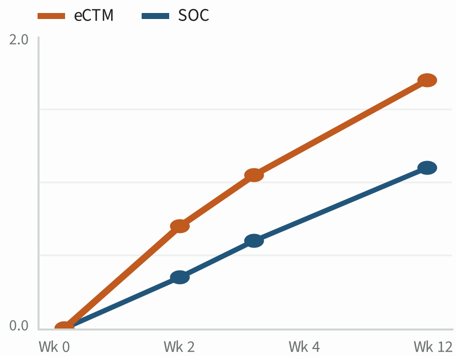
Cumulative improvement in lameness, pain, and swelling — eCTM vs SOC, across 195 conditions.

LAMENESS



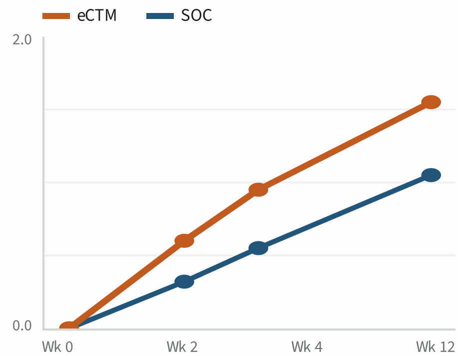
p<0.02 wk 4
p<0.003 wk 12

PAIN



p<0.05 wk 2
p<0.003 wk 12

SWELLING



p<0.03 wk 2
p<0.003 wk 12

2-WEEK SIGNAL

By Week 2, eCTM already showed statistically significant improvement over SOC in pain (p<0.05) and swelling (p<0.03) — a fast-onset effect uncommon in regenerative therapies.

4- TO 12-WEEK SIGNAL

By Week 4, lameness also crossed significance (p<0.02). All three outcomes reached p<0.003 by Week 12. Every eCTM curve was above every SOC curve at every follow-up.

FOLLOW-UP INTEGRITY

98–100% follow-up at every visit for both arms. Panels aggregate all 195 conditions (eCTM 128, SOC 67). The APS-only n=22 comparator subset appears separately in Section 09.

WHAT TO TELL THE VET

The redrawn clinical trajectories. Same story as Section 04 but chart-first — use it if a vet is visual. Point to the Week-2 and Week-4 separation points. **MANUSCRIPT · FIGURE 2 P14**

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Figure 2. p-values from manuscript; curve points redrawn from figure.

SECTION APX A5 · SOC ROSTER & INJECTIONS

What clinicians actually chose — and how injections stacked up

The full SOC product roster and every injection administered across 138 horses.

SOC PRODUCT ROSTER — TABLE 6 (% OF ARM)

Treatment	eCTM arm	SOC arm
Synthetic non-biologic	0.0%	10.6%
› Hyaluronic acid (HA)	0.0%	6.4%
› Polyacrylamide gel (PAAG)	0.0%	4.3%
Biologic — hemoderivative	100%	70.2%
› Autologous protein solution (APS)	0.0%	31.9%
› Platelet-rich plasma (PRP)	0.0%	23.4%
› Autologous conditioned serum (ACS)	0.0%	8.5%
› Platelet-poor plasma (PPP)	0.0%	6.4%
Biologic — regenerative	100%	19.1%
› equicenta® CTM	100%	0.0%
› Amnion allograft	0.0%	8.5%
› Stem cells (bone marrow)	0.0%	8.5%
› Stem cells (adipose)	0.0%	2.1%
› Bone marrow aspirate	0.0%	2.1%
› Exosomes	0.0%	4.3%
Physical therapy	47.3%	40.4%
› Rehabilitation	27.5%	17.0%
› Shockwave	19.8%	23.4%

% of each arm (eCTM n=91, SOC n=47). Sums may exceed 100% (multiple modalities per horse).

INJECTIONS BY TYPE — TABLE 5

Category	eCTM	SOC	Total
Total horses	**91**	**47**	**138**
Total conditions	128	67	195
Primary injections	128	67	195
Repeat SOC injections	0	7	7
Rescue — SOC → SOC	0	2	2
Rescue — SOC → eCTM	0	11	11
Rescue — eCTM → eCTM	9	0	9
Total injections	**148**	**76**	**224**

SIGNIFICANTLY FEWER RE-INJECTIONS

SOC horses required significantly more additional injections than eCTM ($p<0.03$) — driven by 7 protocol-prescribed repeats and 13 rescue injections (11 crossed over to eCTM).

RESCUE DIRECTIONALITY

11 SOC horses crossed over to eCTM as their rescue. 0 eCTM horses crossed over to SOC. eCTM rescues (n=9) went eCTM→eCTM.

SOC WAS NOT CHEAP

APS (32%) and PRP (23%) dominated SOC — the two most expensive point-of-care biologics in equine practice. eCTM is not competing against saline.

WHAT TO TELL THE VET

Tables 5 and 6 — exactly what was injected, and how often. Use it to show the SOC arm was legitimate best-practice care, not a straw man. Names every product in the comparator arm. MANUSCRIPT · TABLES 5 & 6 P13

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Tables 5 and 6. doi:10.3389/fvets.2026.1833933

SECTION APX A6 · SUBGROUPS & HEAD-TO-HEAD

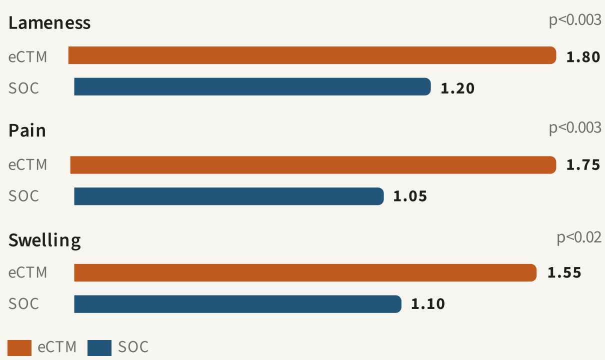
The signal held in joints and in ligaments — and against the strongest SOC

Figure 7 tissue-type subgroup analysis + Suppl Fig S3 head-to-head vs APS (SOC's most-used product).

FIGURE 7 · TISSUE-TYPE SUBGROUP (WK 12 CLINICAL IMPROVEMENT)

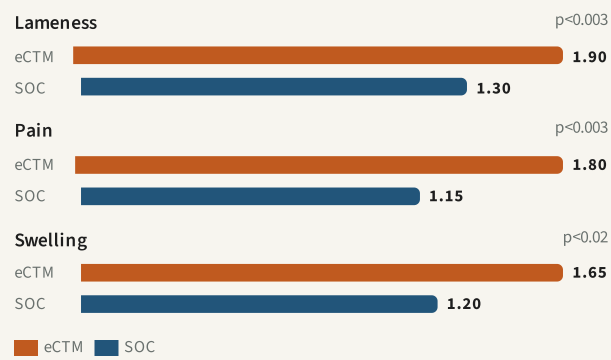
Joint / osteoarthritis cases

n=112 conditions · improvement 0–2 scale



Soft-tissue / ligament cases

n=64 conditions · improvement 0–2 scale



SUPPL FIG S3 · HEAD-TO-HEAD VS APS (SOC'S #1 PRODUCT)

APS IS THE BENCHMARK

APS was SOC's most-used product (31.9%) and one of the most expensive point-of-care biologics in equine practice. eCTM was compared head-to-head against APS in a matched subset.

ECTM WON ON EVERY OUTCOME

In the eCTM (n=39) vs APS (n=22) subset, eCTM was numerically superior on lameness, pain, and swelling at every Wk 12 outcome — and significantly better on the primary composite.

WHY SUBGROUP RESULTS MATTER

Regenerative therapies often shine in one tissue and fail in another. eCTM held its lead in both joints and ligaments, and against SOC's most-used and most-expensive option — evidence the effect is not condition-specific or comparator-of-convenience.

WHAT TO TELL THE VET

Figure 7 (tissue-type subgroups) and the APS head-to-head. Ligaments showed the most pronounced, sustained eCTM advantage. Keep subgroup talk framed as post-hoc/exploratory. **MANUSCRIPT · §3.4.2.6 & FIGURE 7 P17; SUPPL FIG S3 P28**

Source: Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026. Figure 7 and Suppl Fig S3. doi:10.3389/fvets.2026.1833933

SECTION APX B · ADVERSE-EVENT LINE LISTING

Every reported adverse event, by horse and by review type

Supplementary Table S5 — full listing of vet-reported and central-review adverse events.

6	3	13	5
SOC ADVERSE vet-reported	ECTM ADVERSE vet-reported · p<0.04	SOC ADVERSE central review	ECTM ADVERSE central · p<0.002

LINE LISTING — SUPPL TABLE S5

Source	Arm	Severity	Breed	Discipline	Diagnosis	Product	Wk 12 L	Wk 26 VPO
Vet	SOC	Serious	WB	Jumper	LH/RH bilat fetlock arthritis	PAAG	3	euthanized
Vet	SOC	Nonserious	TB	Hunter	RH tarsocrural arthritis	PAAG	3	2
Vet	eCTM	Nonserious	Hackney	Driving	LH stifle arthritis	eCTM	0	4
Vet	eCTM	Nonserious	STDB	Racing	LF SDF tendonitis	eCTM	0	3
Vet	eCTM	Nonserious	QH	Racing	RF DDF tendonitis (insertion)	eCTM	4	2
Central	SOC	Serious	QH	Western	LH chronic active PSD	Fat MSC	1	3
Central	SOC	Serious	WB	Hunter	RF coffin joint arthritis	ACS	0	4
Central	SOC	Serious	TB	Hunter	RH tarsocrural arthritis	PAAG	3	2
Central	SOC	Serious	WB	Jumper	LH/RH fetlock arthritis	PAAG	3	euthanized
Central	SOC	Nonserious	QH	Western	LH chronic active PSD	APS	1	3
Central	SOC	Nonserious	WB	Eventing	RF PSD desmopathy	APS	2	2
Central	SOC	Nonserious	WB	Hunter	LH/RH suspensory branch desmitis	PRP	0	1
Central	SOC	Nonserious	TB	Hunter	RH tarsocrural arthritis	PAAG	3	2
Central	eCTM	Serious	QH	Western	RH tarsocrural arthritis	eCTM	3	2
Central	eCTM	Nonserious	TB	Eventing	RH LSB desmopathy	eCTM	4	2
Central	eCTM	Nonserious	Hackney	Driving	LH stifle arthritis	eCTM	0	4
Central	eCTM	Nonserious	STDB	Racing	LF SDF tendonitis	eCTM	0	3
Central	eCTM	Nonserious	QH	Racing	RF DDF tendonitis	eCTM	4	2

SIGNIFICANTLY FEWER EVENTS ON ECTM

Both vet-reported ($p<0.04$) and central-review ($p<0.002$) adverse events were less frequent in eCTM than SOC. After central review, fewer horses had any reported AE on eCTM ($p<0.04$).

SEVERITY PATTERN

SOC serious events were concentrated in PAAG (3 of 4) and involved chronic joint disease in older or heavier horses. The single eCTM serious event was a tarsocrural arthritis case that resolved with a rescue injection.

WHAT TO TELL THE VET

The full adverse-event line listing (Supplementary Table S5) — every reaction, by horse, both arms. Offer it to a detail-oriented vet: 'here's every event, nothing hidden.' The one euthanized horse was a SOC PAAG case, not eCTM. MANUSCRIPT · SUPPL TABLE S5 P33

Source: Bertone AL et al. Front. Vet. Sci. 2026. Suppl Table S5. L = Wk 12 lameness grade; VPO = Wk 26 vet performance outcome.

SECTION APX C · RESCUE-INJECTION LINE LISTING

Every rescue injection, by limb, diagnosis, and product

Supplementary Table S6 — 22 rescue injections across the trial (hindlimbs 4.6:1 vs forelimbs, p<0.02).

2	11	9	22
SOC → SOC	SOC → ECTM	ECTM → ECTM	TOTAL RESCUES
same-arm rescues	crossover rescues	same-arm rescues	195 conditions

LINE LISTING — SUPPL TABLE S6

Rescue category	Lameness at rescue	Limb	Diagnosis	Product path
SOC → SOC	3	RH	Tarsocrural arthritis	ACS → PAAG
SOC → SOC	2	LF	SDF tendonitis	APS → Fat MSC
SOC → eCTM	4	RH	Hip arthritis	PRP → eCTM
SOC → eCTM	3	LH	Suspensory desmitis	HA + PSGAG → eCTM
SOC → eCTM	2	LH	Suspensory desmitis	Fat MSC → eCTM
SOC → eCTM	3	LH	Stifle arthritis	PRP → eCTM
SOC → eCTM	1	LH	Fetlock arthritis	HA → eCTM
SOC → eCTM	1	LH	Fetlock arthritis	PAAG → eCTM
SOC → eCTM	2	RF	LSB desmitis	PSGAG → eCTM
SOC → eCTM	2	LF	Suspensory desmitis	PRP → eCTM
SOC → eCTM	3	LH	Stifle arthritis	Cultured BM MSC → eCTM
SOC → eCTM	1	LH	Stifle arthritis	PPP → eCTM
SOC → eCTM	1	RH	Stifle arthritis	PPP → eCTM
eCTM → eCTM	3	LH	Collateral ligament desmitis	eCTM
eCTM → eCTM	3	LH	Tarsocrural synovitis	eCTM
eCTM → eCTM	4	RH	Hip arthritis	eCTM
eCTM → eCTM	2	RF	Coffin joint arthritis	eCTM
eCTM → eCTM	3	RH	LSB desmitis	eCTM
eCTM → eCTM	3	LF	Coffin joint arthritis (MRI cartilage damage)	eCTM
eCTM → eCTM	3	RH	Tarsocrural arthritis	eCTM
eCTM → eCTM	4	LF	Carpus arthritis	eCTM
eCTM → eCTM	2	RF	Carpus arthritis	eCTM

DIRECTIONAL FLOW

11 of the 20 arm-comparable rescues crossed from SOC to eCTM. Zero eCTM horses were rescued to SOC. eCTM rescues stayed on-arm.

HINDLIMB DOMINANCE

Hindlimbs were rescued 4.6× more often than forelimbs (p<0.02). SOC arm skewed further toward hindlimb rescue (2.6:1 vs eCTM, p>0.2).

STIFLE & TARSOCRURAL PATTERN

Stifle arthritis (5 rescues), tarsocrural arthritis / synovitis (4), and suspensory desmitis (4) accounted for 13 of 22 rescues. Chronic joint disease drove the rescue caseload on both arms.

WHAT TO TELL THE VET

The rescue line listing (Supplementary Table S6). The headline: 11 SOC horses crossed over to eCTM as rescue, zero eCTM crossed to SOC. Use it to show which way clinicians moved when first-line failed. MANUSCRIPT · §3.4.2.3 P15; SUPPL TABLE S6 P34

Source: Bertone AL et al. Front. Vet. Sci. 2026. Suppl Table S6. Lameness at rescue graded 0–4. PSGAG = polysulfated GAG.

SUPPLEMENTARY MATERIALS · FIGURES S1–S3 · TABLES S1–S6

Supplementary figures & tables

The complete set of supplementary figures and tables reproduced from the peer-reviewed publication — every figure and table on its own page for chairside reference.

IN THIS SECTION

HOW TO USE THIS SECTION

Each page reproduces one published figure or table with its manuscript caption. Flip to any item and use the **What to tell the vet** box to walk through what it shows and why it matters — in plain barn-side language.

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Supplementary Materials. doi:10.3389/fvets.2026.1833933

SUPPL FIGURE S1 · IMAGING & SCORE TRAJECTORIES

Case A vs Case B — imaging & score trajectories

Ultrasound B-mode, Ocean-Floor colorized mapping, 3D terrain, and TRS / MS Core / Quality-Layer scores, Week 0–13.

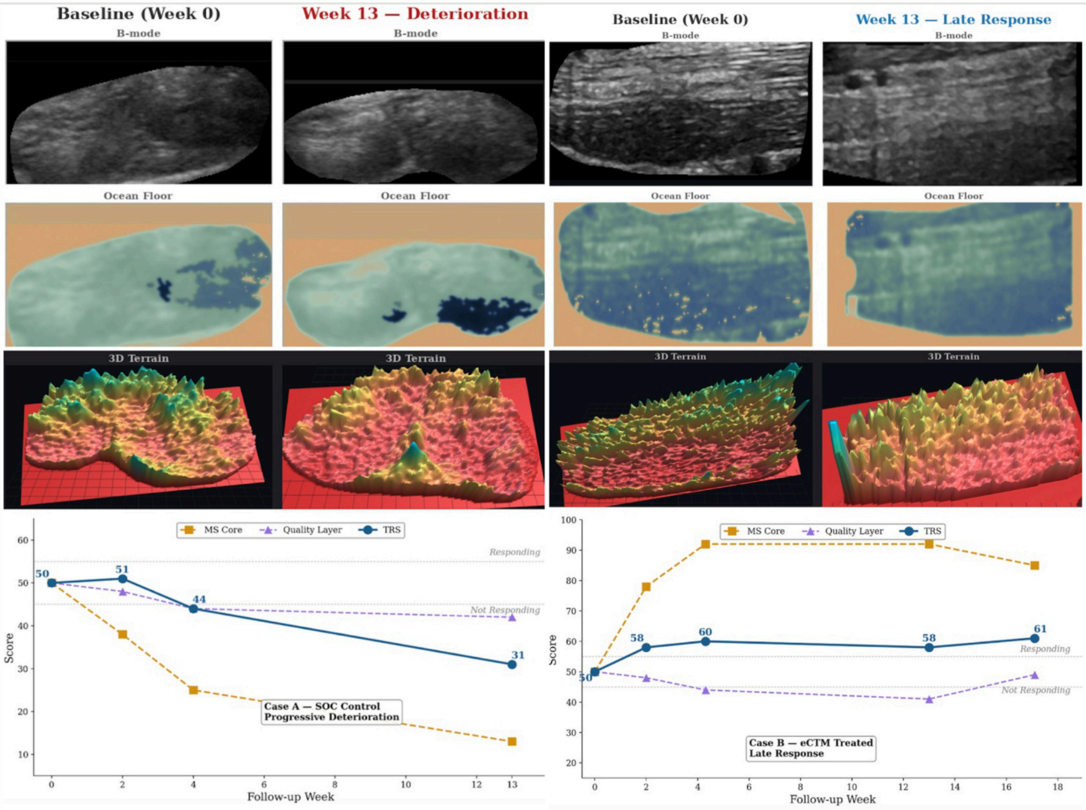


FIGURE CAPTION

Case A (SOC control) — a hind fetlock arthropathy case showing progressive deterioration in ultrasound B-mode, Ocean-Floor colorized mapping, and 3D terrain reconstruction from Baseline (Week 0) to Week 13, with corresponding declines in the tissue-response score (TRS), MS Core score, and Quality-Layer score. **Case B (equicenta® CTM treated)** — a proximal suspensory ligament case showing tissue substrate improvement on all three imaging modalities from Baseline (Week 0) through Week 13, with rising TRS and MS Core scores crossing the 'Responding' threshold and stabilizing at 26-week follow-up.

WHAT TO TELL THE VET

Point to the two halves: the left case is a standard-of-care horse that got **worse** — the tissue map darkens and the scores fall below the 'Responding' line. The right case is an eCTM horse whose tissue actually **rebuilt** — the maps fill in and the TRS/MS Core scores climb above 'Responding' and hold at 26 weeks. This is the imaging you can show an owner: it's not just lameness grades, it's the tissue substrate improving. Flag it as an exploratory imaging case example, not a primary endpoint. MANUSCRIPT · SUPPL FIGURE S1 P26

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Supplementary Figure S1. doi:10.3389/fvets.2026.1833933

SUPPL FIGURE S2 · CONVERGENT TREATMENT-EFFECT SIGNAL

Convergent treatment-effect signal across modalities

Standardized treatment effect (Cohen's d) for TRS and VPO Day 180 in ligaments, tendons, and all soft-tissue structures combined.

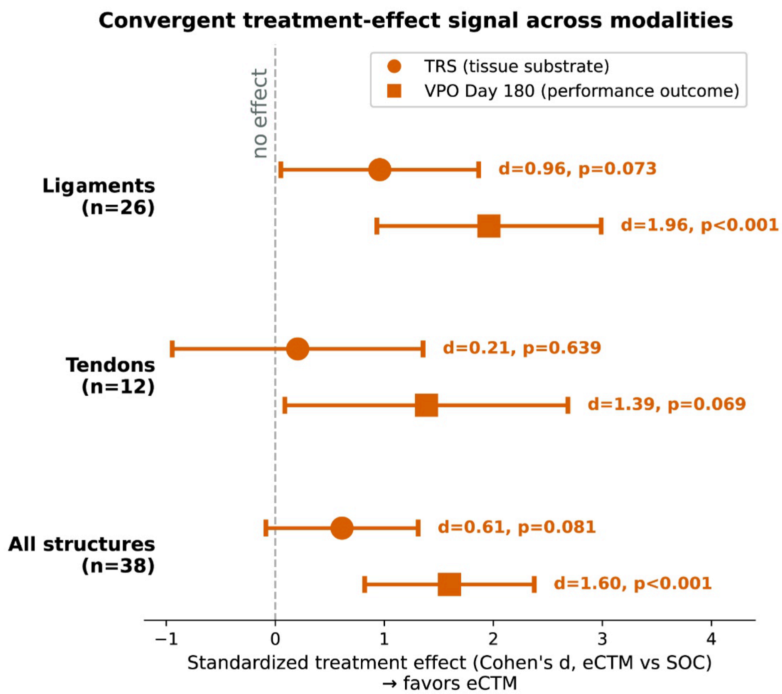


FIGURE CAPTION

Convergent treatment-effect signal across modalities. Standardized treatment effect (Cohen's d, equicenta® CTM vs SOC) for the tissue-substrate score (TRS, circles) and the Veterinary Performance Outcome at Day 180 (VPO, squares), plotted separately for Ligaments (n=26), Tendons (n=12), and All soft-tissue structures combined (n=38). Both modalities point in the direction favoring equicenta® CTM; VPO Day 180 effects are large and statistically significant in ligaments (d=1.96, $p<0.001$) and in all structures combined (d=1.60, $p<0.001$). TRS effect sizes are of similar magnitude but with wider confidence intervals given tissue-level imaging noise.

WHAT TO TELL THE VET

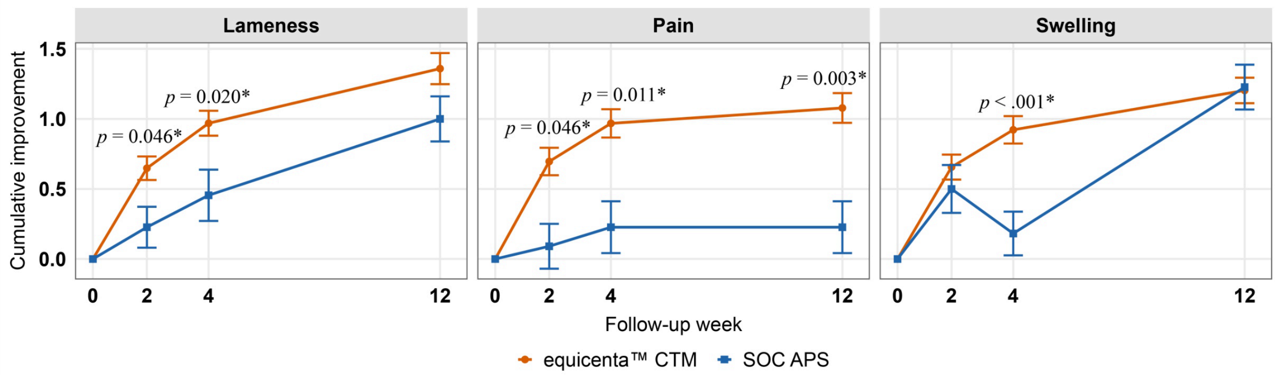
Everything to the **right** of the dashed line favors eCTM. Notice both the imaging score (circles) and the real-world performance outcome (squares) land on the same side — two independent measures agreeing. The performance effect in ligaments is large and significant (d=1.96, $p<0.001$), and across all structures (d=1.60, $p<0.001$). Be conservative on tendons (n=12) — wide bars, not significant. Frame this as: 'the tissue imaging and the horse's actual return-to-work tell the same story.' MANUSCRIPT · SUPPL FIGURE S2 P27

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Supplementary Figure S2. doi:10.3389/fvets.2026.1833933

SUPPL FIGURE S3 · CUMULATIVE IMPROVEMENT 0–12 WK

Cumulative improvement — lameness, pain & swelling (0–12 weeks)

Time-course of cumulative improvement, equicenta® CTM (orange) vs standard-of-care APS (blue), with significance markers at each visit.



	Wk 0	Wk 2	Wk 4	Wk 12
equicenta™ CTM	128	125	128	128
SOC APS	22	22	22	22

	Wk 0	Wk 2	Wk 4	Wk 12
equicenta™ CTM	128	125	126	128
SOC APS	22	22	22	22

	Wk 0	Wk 2	Wk 4	Wk 12
equicenta™ CTM	128	125	128	128
SOC APS	22	22	22	22

FIGURE CAPTION

Cumulative improvement over 12 weeks for lameness, pain, and swelling in horses treated with equicenta® CTM (orange) versus standard-of-care APS (blue). Separation from SOC emerges at Week 2 for lameness ($p=0.046$) and pain ($p=0.046$), and at Week 4 for swelling ($p<0.001$). Improvement continues through Week 12 for lameness ($p=0.020$ at Wk 4) and pain ($p=0.003$ at Wk 12). Sample sizes: equicenta® CTM n=125–128 per visit; SOC APS n=22 per visit. Error bars are standard error of the mean. Asterisks denote statistically significant differences vs SOC at that visit.

WHAT TO TELL THE VET

Orange is eCTM, blue is APS — the single most-used and most-expensive SOC product. eCTM pulls ahead **fast**: lameness and pain separate by Week 2 ($p=0.046$), swelling by Week 4 ($p<0.001$), and the gap holds through Week 12. The message for a vet already reaching for APS: 'this is the head-to-head against your go-to biologic, and eCTM was ahead by the first recheck.' Note this is the 12-week clinical window, separate from the 26-week performance data. MANUSCRIPT · SUPPL FIGURE S3 P28

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Supplementary Figure S3. doi:10.3389/fvets.2026.1833933

SUPPL TABLE S1 · BODY WEIGHT (KG)

Body weight of horses that completed the trial (kg)

Weeks 0 and 12 mean ± SEM for SOC control, equicenta® CTM, and overall.

TABLE S1 · BODY WEIGHT, KG (MEAN ± SEM)

	SOC Control	equicenta® CTM	Overall
Week 0	496 ± 8.1	487 ± 5.0	492 ± 4.2
Week 12	513 ± 7.0	492 ± 4.4	503 ± 3.8
MEAN BY GROUP	**504.5**	**489.5**	—

SOC Control and equicenta® CTM did not statistically differ.

WHAT TO TELL THE VET

This is a **safety and comparability** slide. The two groups weighed the same at the start and stayed stable through 12 weeks (no statistical difference). Use it to answer 'were these comparable horses?' and 'did anything cause weight loss?' — the answer is no. It rules out the horses simply being different sizes or losing condition as an explanation for the results. MANUSCRIPT · SUPPL TABLE S1 P29

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Supplementary Table S1. doi:10.3389/fvets.2026.1833933

SUPPL TABLE S2 · BODY-CONDITION SCORE

Mean body-condition score per horse (weeks 0–12)

Distribution of body-condition scores (3–8) by arm, as counts and percent.

TABLE S2 · MEAN BODY-CONDITION SCORE PER HORSE (WEEKS 0–12)

Body score	SOC n	SOC %	eCTM n	eCTM %
3	1	2.1	0	0.0
4	7	14.9	7	7.7
5	23	48.9	61	67.0
6	10	21.3	18	19.8
7	6	12.8	4	4.4
8	0	0.0	1	1.1
TOTAL	**47**		**91**	

SOC = standard of care. Mean body scores remained within the protocol range (3–8) for SOC and equicenta® CTM. Both equicenta® CTM and SOC had 5 as the most common body score.

WHAT TO TELL THE VET

Same idea as body weight — the horses were **well-matched and healthy**. A body-condition score of 5 (ideal) was the most common in both arms. No horse was under- or over-conditioned in a way that would skew results. Use this if a vet asks whether these were compromised or unusual horses: they were normal, fit performance horses. MANUSCRIPT · SUPPL TABLE S2 P30

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Supplementary Table S2. doi:10.3389/fvets.2026.1833933

SUPPL TABLE S3 · RECRUITMENT MAP

Distribution of practice by recruitment map

Regional breakdown (KY, NC, OH, TX, FL) of horses and conditions by arm and tissue type.

TABLE S3 · DISTRIBUTION OF PRACTICE BY RECRUITMENT MAP — HORSES & CONDITIONS N (%)

Region / arm	Horses	Conditions	Joint	Ligament	Tendon
KENTUCKY	**37 (26.8)**	**53 (27.2)**	**40 (75.5)**	**10 (18.9)**	**3 (5.7)**
SOC	11 (29.7)	17 (32.1)	16 (94.1)	1 (5.9)	0 (0.0)
eCTM	26 (70.3)	36 (67.9)	24 (66.7)	9 (25.0)	3 (8.3)
NORTH CAROLINA	**35 (25.4)**	**44 (22.6)**	**23 (52.3)**	**15 (34.1)**	**6 (13.6)**
SOC	18 (51.4)	21 (47.7)	11 (52.4)	8 (38.1)	2 (9.5)
eCTM	17 (48.6)	23 (52.3)	12 (52.2)	7 (30.4)	4 (17.4)
OHIO	**14 (10.1)**	**21 (10.8)**	**14 (66.7)**	**2 (9.5)**	**5 (23.8)**
SOC	6 (42.9)	9 (42.9)	6 (66.7)	1 (11.1)	2 (22.2)
eCTM	8 (57.1)	12 (57.1)	8 (66.7)	1 (8.3)	3 (25.0)
TEXAS	**10 (7.2)**	**17 (8.7)**	**15 (88.2)**	**0 (0.0)**	**2 (11.8)**
SOC	5 (50.0)	10 (58.8)	10 (100.0)	0 (0.0)	0 (0.0)
eCTM	5 (50.0)	7 (41.2)	5 (71.4)	0 (0.0)	2 (28.6)
FLORIDA	**42 (30.4)**	**60 (30.8)**	**12 (20.0)**	**36 (60.0)**	**12 (20.0)**
SOC	7 (16.7)	10 (16.7)	3 (30.0)	5 (50.0)	2 (20.0)
eCTM	35 (83.3)	50 (83.3)	9 (18.0)	31 (62.0)	10 (20.0)
TOTAL	**138 (100.0)**	**195 (100.0)**	**104 (53.3)**	**63 (32.3)**	**28 (14.4)**
SOC	47 (34.1)	67 (34.4)	46 (68.7)	15 (22.4)	6 (9.0)
eCTM	91 (65.9)	128 (65.6)	58 (45.3)	48 (37.5)	22 (17.2)

Ranked in order of most to least number of horses per Project Veterinarian at the practices in the clinical study.

WHAT TO TELL THE VET

This is your **'this wasn't one clinic'** slide. 138 horses across five states and multiple project veterinarians — Kentucky, North Carolina, Ohio, Texas, and Florida — spanning joints, ligaments, and tendons. If a vet is skeptical it was a single enthusiastic site, show them the geographic and tissue spread: 104 joint, 63 ligament, 28 tendon conditions, real-world case mix. MANUSCRIPT · SUPPL TABLE S3 P31

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Supplementary Table S3. doi:10.3389/fvets.2026.1833933

SUPPL TABLE S4 · PRIMARY CONDITIONS / HORSE

Distribution of primary conditions per horse

Single-site, multi-condition, and multi-limb case-mix by treatment arm (SOC vs eCTM).

TABLE S4 · DISTRIBUTION OF PRIMARY CONDITIONS PER HORSE

A. Primary conditions*	SOC n (n=47)	SOC %	eCTM n (n=91)	eCTM %
1 condition (single site)	30	63.8%	58	63.7%
2 conditions in one lame limb	1	2.1%	11	12.1%
1 condition in two lame limbs	14	29.8%	19	20.9%
3 conditions	2	4.2%	3	3.3%
TOTAL HORSES WITH >1 CONDITION	**17**	**36.2%**	**33**	**36.3%**
TOTAL HORSES	**47**	**100.0%**	**91**	**100.0%**

*There was no significant difference between groups in distribution of primary conditions.

WHAT TO TELL THE VET

Another **comparability** slide, and a real-world one. About a third of horses in **both** arms had more than one problem — multi-site, multi-limb, the complicated cases you actually see, not cherry-picked single lesions. And the split was statistically identical between groups, so eCTM wasn't handed the easy cases. Use it to say: 'these were real, often multi-condition horses, evenly distributed.'

MANUSCRIPT · SUPPL TABLE S4 P32

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Supplementary Table S4. doi:10.3389/fvets.2026.1833933

SUPPL TABLE S5 · ADVERSE EVENTS — ALL INJECTIONS

Adverse events by horse and by report

Veterinarian and central-review scoring, severity, breeds, disciplines, and diagnoses; p-values for eCTM vs SOC.

TABLE S5 · ADVERSE EVENTS BY HORSE AND BY REPORT — ALL INJECTIONS

Group	Severity	Rpt #	Horse #	Practice	Product	Breed	Discipline	Diagnosis	Signs	Wk12 L	Wk26 VPO
VETERINARIAN SCORES											
SOC	**Serious**	2	1	P2	PAAG	WB	Jumper	LH/RH bilateral hind fetlock arthritis	Lameness, swelling	3	euthanized
SOC	Nonserious	4	1	P6	PAAG	TB	Hunter	RH tarsocrural arthritis	Marked effusion, lameness	3	2
eCTM	Serious	0	0	none	none	none	none	none	none	none	none
eCTM	Nonserious	3	3	P1	eCTM	Hackey	Driving	LH stifle arthritis	Lameness, swelling	0	4
eCTM				P3	eCTM	STDB	Racing	LF SDF tendonitis	Lameness	0	3
eCTM				P8	eCTM	QH	Racing	RF DDF tendonitis (insertion)		4	2
Total vet · SOC	**Serious 6**		**Horse 2**								
Total vet · eCTM	**3***		**Horse 3**								
CENTRAL REVIEW											
SOC	**Serious**	9	4†	P2	Fat MSC	QH	Western	LH chronic active PSD	Lame, swollen, painful	1	3
SOC				P6	ACS	WB	Hunter	RF coffin joint arthritis		0	4
SOC				P6	PAAG	TB	Hunter	RH tarsocrural arthritis		3	2
SOC				P2	PAAG	WB	Jumper	LH/RH fetlock arthritis		3	euthanized
SOC	Nonserious	4	4	P2	APS	QH	Western	LH chronic active PSD	Lame	1	3
SOC				P2	APS	WB	Eventing	RF PSD desmopathy	Swelling	2	2
SOC				P7	PRP	WB	Hunter	LH/RH suspensory branch desmitis	Swelling	0	1
SOC				P6	PAAG	TB	Hunter	RH tarsocrural arthritis	Effusion	3	2
eCTM	**Serious**	1	1	P2	eCTM	QH	Western	RH tarsocrural arthritis	Lame, effusion	3	2
eCTM	Nonserious	4	4†	P2	eCTM	TB	Eventing	RH LSB desmopathy	Swelling	4	2
eCTM				P1	eCTM	Hackey	Driving	LH stifle arthritis	Lame	0	4
eCTM				P3	eCTM	STDB	Racing	LF SDF tendonitis	Swelling	0	3
eCTM				P8	eCTM	QH	Racing	RF DDF tendonitis (insertion)	Lameness	4	2
Total central · SOC	**13**		**Horse 8**								
Total central · eCTM	**5****		**Horse 5****								

*Reported adverse reactions by the veterinarian (p<0.04) and central review (p<0.002) were less frequent in eCTM than eSOC. **After central review, fewer horses had an adverse reaction reported in eCTM than SOC (p<0.04). †One horse was a rescue injection. PAAG=polyacrylamide gel; WB=Warmblood; TB=Thoroughbred; QH=Quarter Horse; STDB=Standardbred; MSC=mesenchymal stromal cell; ACS=autologous conditioned serum; PRP=platelet-rich plasma; APS=autologous protein solution. DDF=deep flexor tendon; SDF=superficial flexor tendon; LSB=lateral suspensory branch; PSD=proximal suspensory desmitis.

WHAT TO TELL THE VET

This is the **safety** table — every reported reaction, by horse, in both arms. The headline: **fewer** reactions on eCTM, and it's statistically significant (vet-reported p<0.04, central review p<0.002). Note the single SOC serious case that was euthanized was a PAAG fetlock horse — not eCTM. eCTM had zero serious vet-reported reactions. Stay honest: absolute safety isn't 'established,' this is one trial — but the observed reaction profile favored eCTM on every count. MANUSCRIPT · SUPPL TABLE S5 P33

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Supplementary Table S5. doi:10.3389/fvets.2026.1833933

SUPPL TABLE S6 · RESCUE INJECTIONS

Rescue injections at the target site (per protocol)

Details of SOC→SOC, SOC→eCTM crossover, and eCTM→eCTM rescues; forelimb vs hindlimb odds ratio.

TABLE S6 · RESCUE INJECTIONS AT THE TARGET SITE (PER PROTOCOL)

Rescue injections	Lameness at rescue	Limb	Diagnosis	Product
RESCUE — SOC → SOC				
3	3	RH	Tarsocrural arthritis	ACS to PAAG
2	2	LF	SDF tendonitis	APS to Fat MSC
RESCUE — SOC → ECTM (CROSSOVER)				
4	4	RH	Hip arthritis	PRP
3	3	LH	Suspensory desmitis	HA and PSGAG
2	2	LH	Suspensory desmitis	Fat MSC
3	3	LH	Stifle arthritis	PRP
1	1	LH	Fetlock arthritis	HA
1	1	LH	Fetlock arthritis	PAAG
2	2	RF	LSB desmitis	PSGAG
2	2	LF	Suspensory desmitis	PRP
3	3	LH	Stifle arthritis	Cultured bone marrow MSC
1	1	LH	Stifle arthritis	PPP
1	1	RH	Stifle arthritis	PPP
RESCUE — ECTM → ECTM				
3	3	LH	Collateral ligament desmitis	eCTM
3	3	LH	Tarsocrural synovitis	eCTM
4	4	RH	Hip arthritis	eCTM
2	2	RF	Coffin joint arthritis	eCTM
3	3	RH	LSB desmitis	eCTM
3	3	LF	Coffin joint arthritis (cartilage damage on MRI)	eCTM
3	3	RH	Tarsocrural arthritis	eCTM
4	4	LF	Carpus arthritis	eCTM
2	2	RF	Carpus arthritis	eCTM

LSB = lateral suspensory branch; ACS = autologous conditioned serum; PAAG = polyacrylamide gel; APS = autologous protein solution; PRP = platelet-rich plasma; HA = hyaluronic acid; PSGAG = polysulfated glycosaminoglycan; PPP = platelet-poor plasma; MSC = mesenchymal stromal cells. The odds of hindlimbs (H) having rescue injections compared to forelimbs was 4.6:1 ($p<0.02$); the odds of SOC having more hindlimbs rescued than eCTM was 2.6:1 but did not statistically differ ($p>0.2$).

WHAT TO TELL THE VET

'Rescue' means the first treatment didn't hold and the horse needed re-injection. The story in the columns: **11 SOC horses crossed over to eCTM** as their rescue — and **zero** eCTM horses were rescued with SOC. When SOC failed, clinicians reached for eCTM; the reverse never happened. Hindlimbs needed rescue 4.6× more than forelimbs ($p<0.02$) regardless of arm — a limb-load point, not a product point. Frame it as: 'the crossover only went one direction.'

Source: Bertone AL, Davern A, Sutter WW, et al. Front. Vet. Sci. 2026. Supplementary Table S6. doi:10.3389/fvets.2026.1833933

Evidence from the peer-reviewed record.

This booklet summarizes the peer-reviewed Bertone et al. 2026 controlled trial of equicenta® CTM, pairing manuscript-aligned statistics with the study's own published figures.

PUBLICATION	Bertone AL, Davern AJ, Sutter WW, et al. Front. Vet. Sci. 2026
DOI	10.3389/fvets.2026.1833933
JOURNAL	Frontiers in Veterinary Science · Regenerative & Sports Medicine
TRIAL IDS	IACUC ETCR-2024-0329 · AVMA VCT #26005934
PRODUCT STATUS	Investigational — not FDA CVM approved

equicenta® CTM

Equine Performance Labs · Austin, Texas · equineperformancelabs.com

Read the paper: doi:10.3389/fvets.2026.1833933

ISO 13485-ALIGNED

Manufacturing quality management system

IMPORTANT INFORMATION

Important information: equicenta® CTM is under investigation. It has not been approved by the U.S. FDA Center for Veterinary Medicine, and the safety and effectiveness of the product for the uses described have not been established. Results are from a single prospective controlled clinical study in performance horses; individual outcomes may vary. Not for human use. Not for use in food-producing animals. Source: Bertone AL, Davern AJ, Sutter WW, et al. Cryopreserved umbilical allograft (equicenta® CTM) improves clinical outcomes compared with clinician-selected standard care in performance horses: a pragmatic prospective controlled trial. Front. Vet. Sci. 2026. doi:10.3389/fvets.2026.1833933. IACUC ETCR-2024-0329 · AVMA VCT #26005934.